AIM-418-1
AIM-418-2
AIM-418-3
VIBRATING MACHINE

Aimil Ltd.
Instrumentation & Technologies

Version

2.2

Aimil Ltd.
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AIM - 418-1
AIM- 418-2
AIM-418-3

VIBRATING MACHINE

Operating Instructions

Vibrating Machine

The equipment comprises of the following :

1.	Vibrating Machine consisting of :	
	i. Frame	1 No.
	ii. Vibrator mounted on coiled springs fitted with an eccentric shaft & a pulley	1 No.
	iii. Motor fitted with a pulley	1 No.
	iv. Belt	1 No.
	v. Belt Guard	1 No.
	vi. Digital Timer	1 No.
2.	ISI marked Cube Mould (ISI marked), 70.6mm with base plate	1 No.
3.	Poking Rod	1 No.
4.	Lubricating oil	1 No.
5.	Spanner	1 No.
6.	Calibration Certificate for R.P.M. from National Council for Cement and Building Materials or from an NABL accredited Lab.	1 No.
7.	Operating Instructions with General Assembly Drawing	1 No.

Note : AIM-418-1 - Vibrating machine with NCCBM certificate for RPM.
AIM-418-2 - Vibrating machine with NABL accredited lab certificate for RPM.
AIM-418-3 - Vibrating machine without Calibration Certificate

CAUTION: BEFORE OPERATING THE MACHINE ALWAYS USE 3 IN 1 SEWING MACHINE OIL OR EQUIVALENT.

Operating Instructions
Vibrating Machine

Introduction

The machine is designed to meet the requirements of IS:10080 and IS:4031 for the preparation of mortar cubes for the determination of compression strength of ordinary and rapid hardening portland cement, low heat portland cement, portland blast furnace cement and high alumina cement. The R.P.M. is 12000 ± 400. It is fitted with electric motor for operation on a 220 ± 10 Volts, 50 Hz, Single phase, AC supply.

Description

The equipment is illustrated in the General Assembly Drawing. The numbers given against the components of the machine in the description below pertain to this General Assembly Drawing.

It consists of a vibrator (13) mounted on springs (12) to carry the cube mould (5) and a revolving eccentric shaft. By means of a balance weight beneath the base plate attached rigidly to the frame, the centre of gravity of the whole machine, including the cube and the mould, is brought either to the centre of the eccentric shaft or within a distance of 25mm below it. Due to this, the revolving eccentric arrangement imparts an equal circular motion to all the parts of the machine and the mould. The motion thus attained is equivalent to equal vertical and horizontal simple harmonic vibrations, 90 degrees out of phase. The minimum running speed of the machine is well above its natural frequency on its supporting springs, hence the amplitude of vibration is independent of the speed. The drive is provided by means of an endless belt running between a crowned pulley on the motor (7) and a crowned pulley on the vibrator. One belt cover (6)is clamped on the base to cover the belt for safety. One Hopper (4) is clamped on the top of mould and one screw with a handle is provided for clamping the mould. The electric motor is mounted on the base with the help of rubber vibrating rubber pads (02) & motor adjustment bolts (01). The main base is provided with levelling screws (14) with clamping nuts. The Tension Spring (3) is mounted on the main base to give tension to the vibrator. An ON-OFF switch (10) & a digital timer (09) for starting or stopping the machine is fixed on the frame of the machine.

Operation

Caution:

1. The vibrator is clamped to the main base with the help of an L shaped strip and bolts while packing the machine for despatch.

2. Remove the clamp before starting the machine.

Moulding and Curing of Specimens

Take standard sand of the white variety which 100% passes the 2mm IS Sieve and is 100% retained on the 90 micron IS Sieve (Ennore Sand has been standardised. Refer to IS:650).

For each cube, weigh the cement, sand and water separately as follows :

Cement	200 g	
Standard Sand	600 g	
Water	$\left[\frac{P}{4}+3.0\right]$	percent of combined weight of cement and sand, where P is the percentage of water required to produce a paste of standard consistency.

Note : ***For determining Normal Consistency of Cement AIM-393, Vicat Apparatus is recommended. (Can be supplied at an extra cost).***

Mix weighed quantities of cement and dry sand with a Trowel, AIM-428 (can be supplied at an extra cost) for one minute. Then add the calculated quantity of water and mix the three ingredients thoroughly until the mixture attains a uniform colour. If a uniform colour is not obtained within four minutes reject the mixture and prepare another mixture. Gauging time is very important, it should not exceed 4 minutes. A clean appliance should be used for gauging.

Setting Up

Level the machine with the help of the levelling screws (14). Set the tension of the belt, align the motor pulley and crowned pulley of vibrator by using the adjustment of bolts (2 &1) & the tension spring (3) such that the mould with base can be kept vertically over the springs. Remove the hopper (4) and cube mould (5) by unscrewing the nuts and clamping screw for the mould. Apply grease lightly to all the four inside faces of the mould and the base plate. Mount the mould with base plate on the top of the vibrator under the hopper (4). Tightly fit the mould and hopper with the nut and side screws provided. The Hopper should not be removed until the completion of the vibration period.

Adjustments

1. Adjust the base of the mould so that it rests vertically over the four springs. This can be done by moving the spring (4) either upwards or downwards.

2. Check up the tension of the belt. It should be tight without slackness.

3. The bed of the machine shall be horizontal. This can be done by levelling screws provided.

4. The wheel of the motor and the eccentric shaft must lie in the same line and in the same plane (5).

5. Check the alignment of both the pulleys.

6. Check the motor connection before switching on. If during the course of running, the belt is vibrating, then the following adjustments are suggested :

   a. the belt comes off towards outside, adjustment bolt (1) of the motor bed plate should be tightened further.

   b. if the belt comes towards inside and touches the bearing disc, bolt of the motor bed plate should be tightened.

   c. to get more tension in the belt, the spring (3) should be tightened.

   d. if there is a whistling noise from the belt drive, then the plate carrying the spring in front of the belt should be adjusted either upwards or downwards and locked in position.

Working of Timer

Working of timer is very simple and easy to do. There are 3 keys in the front panel for setting the time required for vibration. One additional push switch for **RESET** is used to reset the timer only.

Operating the Vibrating Machine

Immediately after mixing, transfer the cement paste through the hopper into the assembled mould placed in position on the table of the vibration machine. Prod the mortar with a Poking Rod 20 times in about 8 seconds to ensure the elimination of any entrained air and honey combing that may be present. Place the remaining quantity of mortar in the hopper of the mould and prod it again as specified for the first layer and then compact the specimen by vibration.

Start the machine and vibrate for two minutes. At the end of two minutes of vibration, remove the mould with the base plate from the machine. Finish the top surface of the cube in the mould by smoothening the surface with the blade of a trowel and immediately keep it in a humid atmosphere for 24 hours. After that period, demould and submerge the specimen in clean, fresh water and leave it there until it is taken out just prior to breaking. Change the water every 7 days. Maintain the water at a temperature of 27°C ± 2°C.

Testing Compressive Strength

Test the cubes for compressive strength at the end of 1 day, 3 days and 28 days (see the table) which periods shall be reckoned from the completion of vibration.

Test three cubes every time and take the average of the three as the compressive strength.

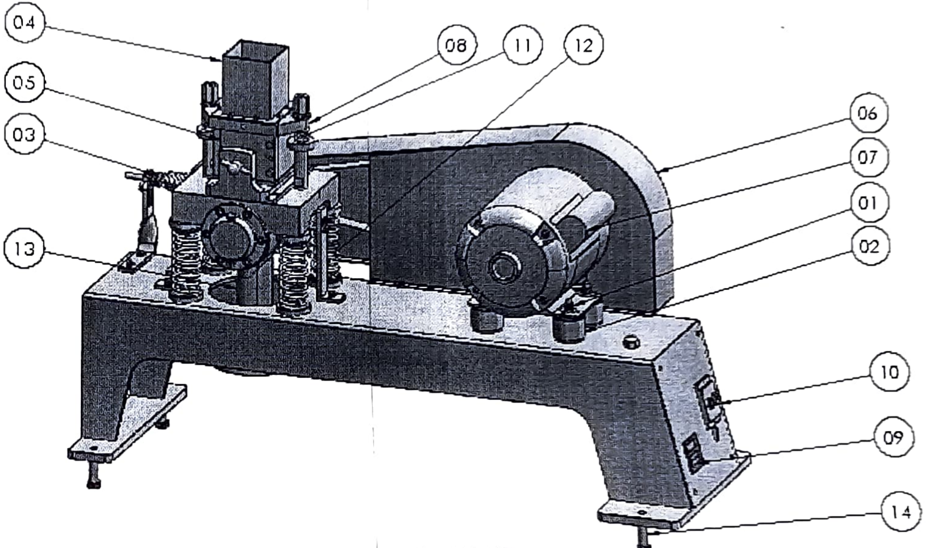
The cubes shall be tested on their sides without any packing between the cubes and the steel platens of the testing machine. Apply load steadily and uniformly at a rate of 350 kgf/cm^2/min. The compressive strength shall be calculated from the crushing load and the average area over which the load is applied.

Compressive Strength of Various Types of Portland Cements at Different Periods of Curing

Age of Test	Ordinary Portland Cement kgf/cm²	Rapid Hardening Portland Cement kgf/cm²	Low Head Portland Cement kgf/cm²
24 hrs. ± 30 min. (Compressive Strength not less than)	-	160	-
72 hrs. ± 1 hour (Compressive Strength not less than)	160	275	100
160 hrs ± 2 hours (Compressive Strength not less than)	220	-	160
672 hrs. ± 4 hours (Compressive Strength not less than)	-	-	350

Maintenance

1. Keep the apparatus clean.
2. Keep all the four faces of the cube mould and the base plate lightly greased.
3. Before operation of the machine, please lubricate the eccentric shaft at the position marked by the indicating arrow by removing the Oil Filler Plug.
4. Lubricate the machine with thin oil at frequent intervals.
5. Before Operating the Machine always use 3 in1 Sewing Machine Oil or Equivalent.



GA DRAWING
AIM - 418-T, VIBRATION MACHINE

S. No.	DESCRIPTION
01	MOTOR ADJUSTMENT BOLT
02	RUBBER PAD
03	TENSION SPRING
04	HOPPER
05	CUBE MOULD (70.6mm)
06	BELT COVER
07	MOTOR
08	HOPPER BASE PLATE

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S. No.	DESCRIPTION
09	DIGITAL TIMER
10	ON/OFF SWITCH
11	SPRING FOR FITTING MOULD
12	SUPPORTING SPRING
13	VIBRATOR
14	LEVELLING SCREW